

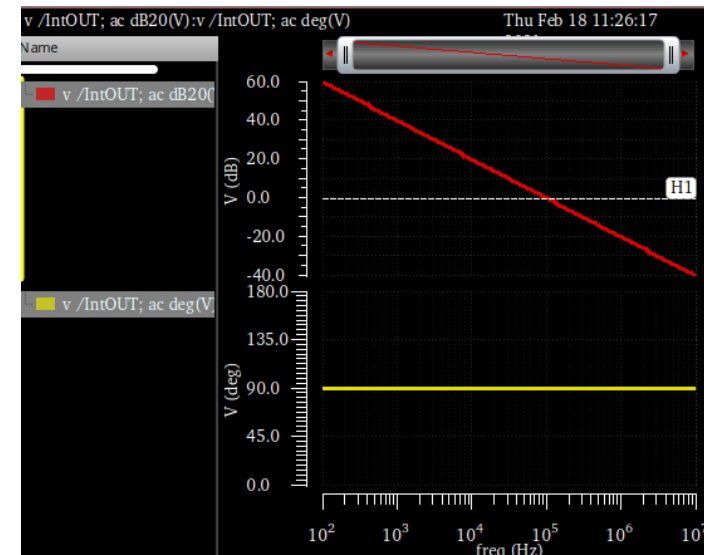
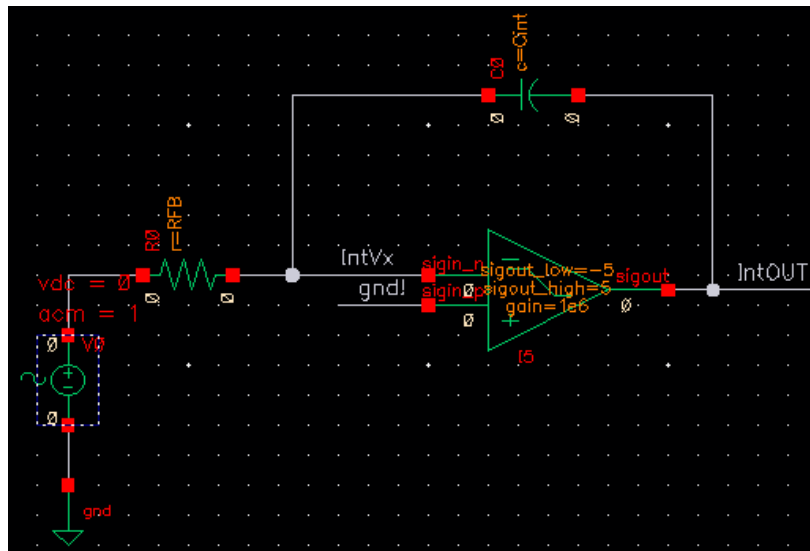
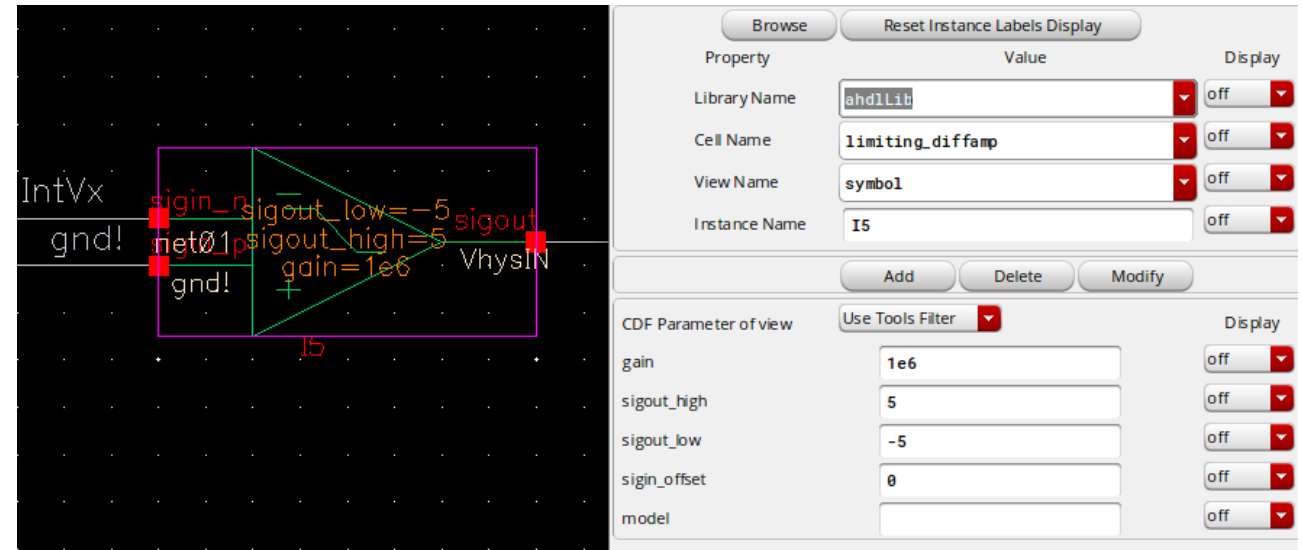
ET4382: Power Conversion Techniques

Tutorial-2: Hysteresis PWM Amplifier

17th February, 2022

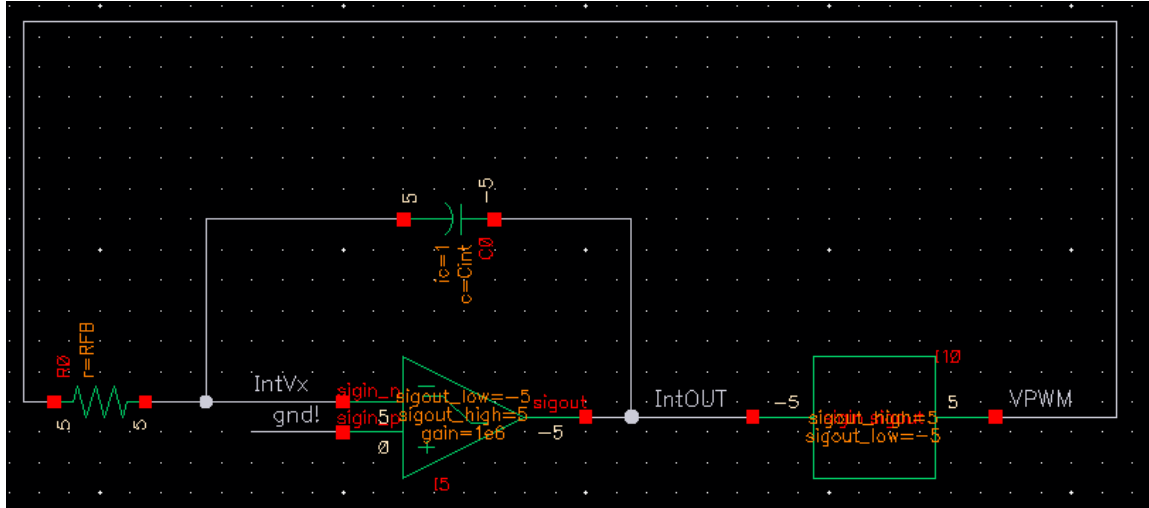
Creating an Active-RC Based Integrator

- An active RC-based Integrator uses a RC network around a high gain amplifier to create a s-domain transfer:
 - $V_{out}(f) = 1/(2\pi RCf) * V_{in}(f)$
- Use an ideal Amplifier (a high gain limiting diffamp).
- Build the active-RC integrator as shown. You may parameterise the RC values.
- Run an AC analysis to check the Bode plot and confirm the UGB.



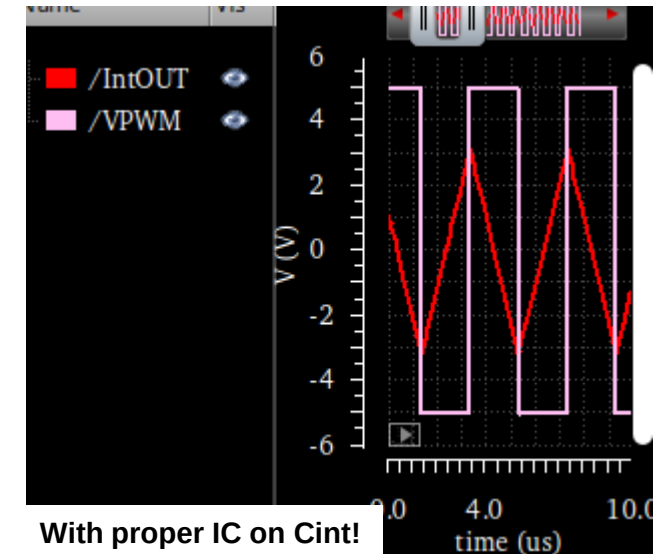
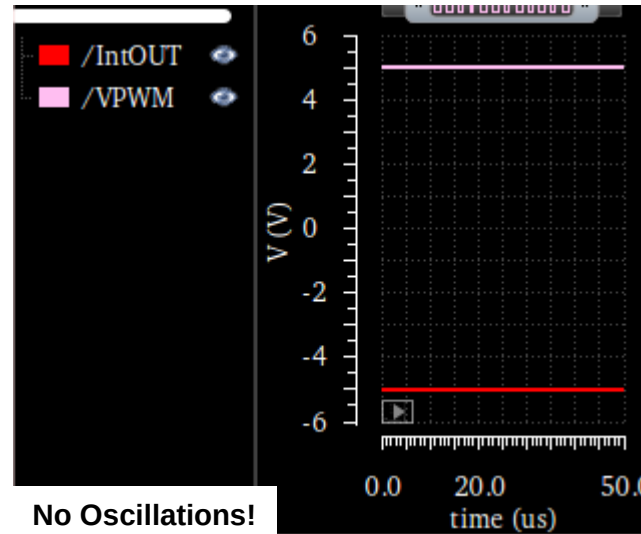
Creating a Hysteresis Loop

- A hysteresis loop is simply an oscillator, made by adding a delay in closed loop around an integrator.



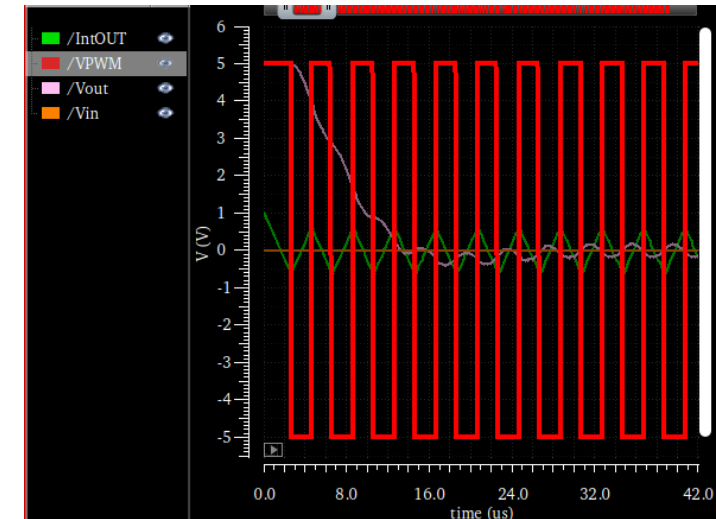
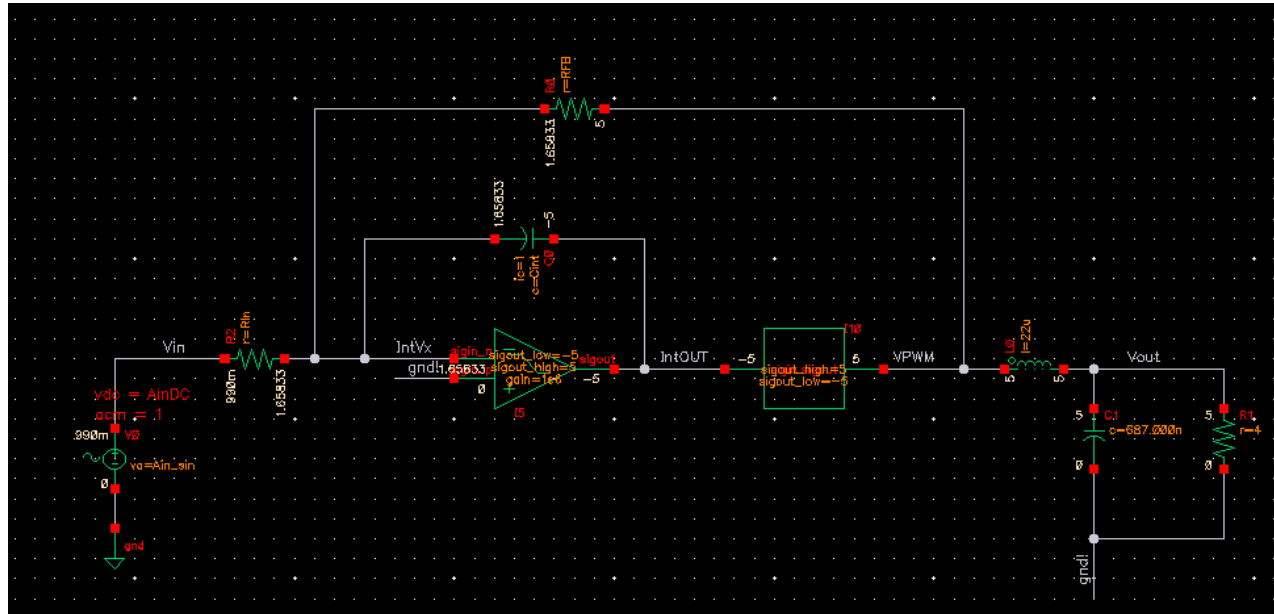
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hyst_state_init			off
sigout_high	5		off
sigout_low	-5		off
sigtrig_low	-vhys		off
sigtrig_high	vhys		off
tdel	tdel_hys		off
trise			off
tfall			off
model			off

- Use a hysteresis block (ahdLib) using parameters as shown and close the loop around the integrator.
- This hysteresis block is basically comparing with 0 with a hysteresis of $\pm v_{hys}$ and a delay of $tdel_hys$. Start by setting $v_{hys} = 0$, and $tdel_hys = 1\mu$.
- Note:** The hysteresis block is very ideal and runs on edge-triggered logic. As seen in the figure the DC point before and after the block is -5 and +5 respectively. Where in, it should have been $V(VPWM) = -5$, since $V(IntOUT) < 0$. This may not be rectified in a transient simulation if not set up properly and you won't notice any oscillations.
- To avoid this scenario use an initial condition of 1V across the capacitor.

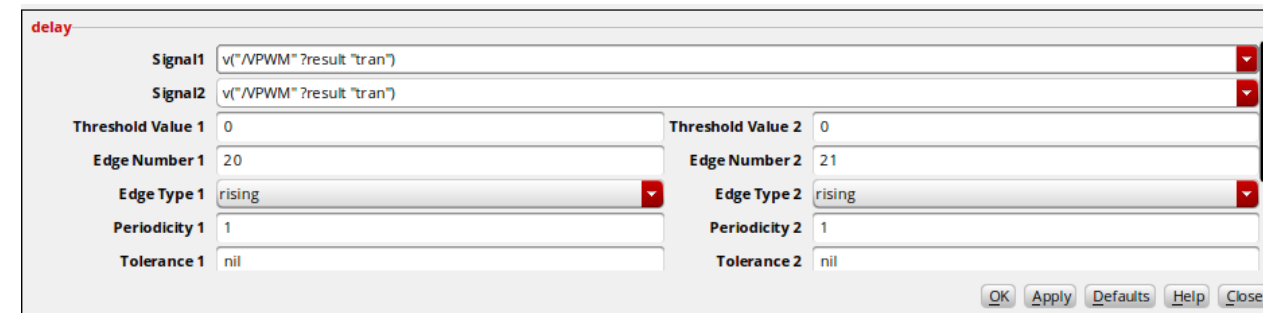


Creating a Class-D Hysteresis Amplifier.

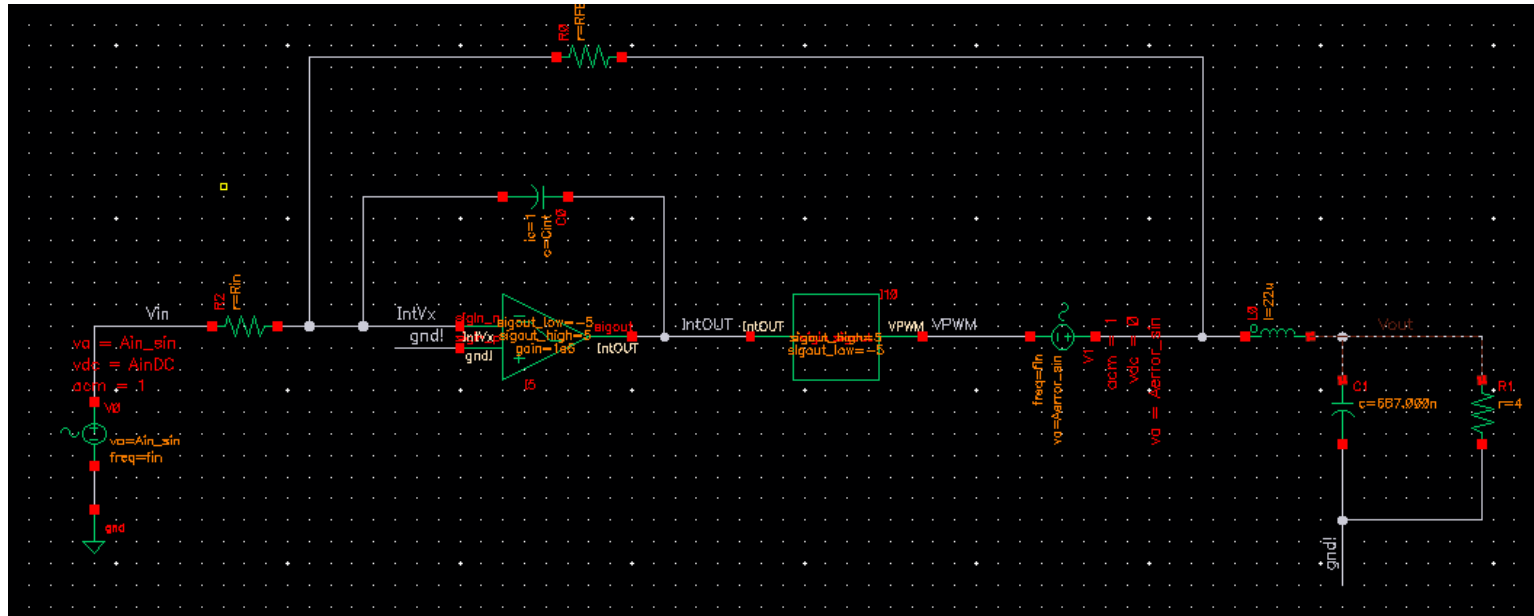
- Convert it to closed loop class-D amplifier by adding a separate input resistor to provide the input.
- Set the amplifier for a gain of 5x keep the feedback resistor as 100k. **Note:** The integrator UGB is now defined using R_{in} and C_{int} .
- 1FS is therefore defined as 5V at output and 1V at input.



- Provide an input DC bias (0V) + Sine Amplitude (0V) and ensure that the PWM frequency remains the same.
- Use the calculator to read the PWM frequency more conveniently.
- Rt+Click on PWM Node and send to calculator and use the delay function. Set it up as shown. The edge numbers define the successive edges across which $f_{pwm} = 1/t_{pwm}$ is calculated.
- Ensure that the PWM frequency is calculated once the output has settled sufficiently. (So run for a sufficient number of cycles.)



Adding an Error Voltage Source



- To check the suppression of the non-idealities of the PWM convertor an error voltage source can be added at the end, before the feedback.
- Try giving it a small amplitude and have a look at the FFT. (Keep input amplitude 0 in this case)
- You may also try remove the hysteresis block and plot the transfer function before and after the error source.